

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'A' [MCQ] [11×1=11]

- Number of electrons required to balance the following equation.
 $MnO_4^{--} + 4H^+ \rightarrow MnO_2 + 2H_2O$ is
 a) 5 b) 4 c) 3 d) 2
- Which of the following reaction is non redox reaction?
 a) $SO_2 + H_2S \rightarrow 2H_2O + 3S$ b) $2Na + O_2 \rightarrow Na_2O_2$
 c) $Na_2O + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$ d) $4KClO_3 \rightarrow 3KClO_4 + KCl$
- A solution of SO_2 in water reacts with H_2S to precipitate sulphur. SO_2 is acting as
 a) reducing agent b) oxidizing agent
 c) catalytic nature d) an acid
- Strongest reducing agent among the following is
 a) $HClO$ b) $HClO_2$
 c) $HClO_3$ d) $HClO_4$
- One liter of an unknown gas weighs 1.25 gm at NTP. Which of the following gas illustrates above data?
 a) NO_2 b) N_2
 c) CO_2 d) O_2
- The vapour density of X is four times that of Y. The molecular mass of X is M, then the value of Y is
 a) 4M b) M/2 c) M d) M/4
- The volume of a gas at STP is 133 ml. What is the volume at 273°C and 2 atmospheres?
 a) 133 ml b) 126 ml c) 563 ml d) 66.5 ml

8. The correct order of size of I , I^+ and I^- is
 a) $I > I^+ > I^-$
 c) $I^+ > I > I^-$

b) $I > I^- > I^+$

d) $I^- > I > I^+$

9. Which of the following is expected to have highest second ionization energy?
 a) $1s^2 2s^2$ *13C*
 c) $1s^2 2s^2 2p^1$

b) $1s^2 2s^2 2p^4$

d) $1s^2 2s^2 2p^3$

10. Vital force theory of the origin of organic compounds was discarded by
 a) Kolbe's synthesis
 c) Wohler synthesis

b) Berthelot synthesis

d) Haber's synthesis

11. The property of catenation is shown by the element
 a) Nitrogen
 c) Carbon

b) Sulphur

d) Oxygen

GROUP 'B' [8×5=40]

Short Questions.

- a) Balance the following redox reaction either by oxidation number method or by ion-electron method.
 $Mg + HNO_3 \rightarrow Mg(NO_3)_2 + N_2O + H_2O$
 b) Identify the number of acidic and oxidizing HNO_3 in above reaction. [4+1]
- a) Define oxidation number with example. [1]
 b) What is the oxidation number of Sulphur in Caro's acid (H_2SO_5)? [1]
 c) The reaction occurs as follows, $HCl + NaOH \rightarrow NaCl + H_2O$; Is the reaction redox or not? Why? [1]
 d) Give one example each of the compound having maximum and minimum oxidation number for Sulphur and Chlorine. [2]
- a) Define electron affinity. [1]
 b) Compare the electron affinity of Fluorine with chlorine. [2]
 c) Differentiate between electron affinity and electronegativity. [2]
- a) Show your familiarity with differentiating electron classification. [3]
 b) Compare the ionic size of F^- , Ne & Na^+ with reason. [2]
- a) Write a short note on "Homologous series & its characteristics." [3]
 b) Write differences between organic and inorganic compounds. [2]

6. a) Derive ideal gas equation. [2]
 b) Find the value of R in CGS unit. [1]
 c) Derive the relationship between Partial pressure, mole fraction and total pressure of gaseous mixture. [2]
 7. a) State Dalton's Law of partial pressure. Write down its application. [2]
 b) A vessel contains 12g of an ideal gas at $t^{\circ}\text{C}$ temperature and 1 atm pressure. When the temperature is increased by 10°C at the same volume, the pressure is increased by 10%. Calculate the volume and initial temperature. (Molecular mass of gas = 120) [3]
 8. a) Define absolute zero temperature. Explain it with suitable graph. [2]
 b) A helium filled balloon is designed to rise to a height of 50 km at which it will be fully inflated and have the volume of one hundred thousand litres. At this altitude the temperature is -5°C and atmospheric pressure is 1.52 mm of Hg. How many litres of helium at NTP will be required to fully inflate the balloon and what will be its weight? [3]

GROUP 'C' [8×3= 24]

Long Questions

9. a) State and explain Graham's Law of diffusion. [3]
 b) What are the differences between diffusion and effusion of gas? [2]
 c) The mass of 525 cc of a gaseous compound at 28°C and 730 mm of Hg pressure was found to be 0.9g. What will be the volume of 2g of the gas at 30°C and 760 mm of Hg pressure? [3]
 10. a) Balance the following equation either by oxidation number method or by ion-electron method. [3+3]
 i) $\text{CrO}_4^{2-} + \text{HSnO}_2^{-} \rightarrow \text{CrO}_2 + \text{HSnO}_3^{-}$ (in basic medium)
 ii) $\text{P} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{PH}_3 + \text{NaH}_2\text{PO}_2$ [2]
 b) Define oxidant and reductant giving suitable example. [1]
 11. a) Define functional group with example. [2]
 b) Write the characteristics of functional group. [1]
 c) Give the reason for the preparation of sodium extract for detection of heteroelement. [2]
 d) How can you detect nitrogen as heteroelement? Explain it with necessary reaction. [3]

Best of Luck!

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$$\text{CrO}_4^{2-} + \text{HSnO}_2^- \rightarrow \text{CrO}_2^- + \text{HSnO}_3^- \text{ (in basic medium)}$$
 - i) $\text{P} + \text{NaOH} + \text{H}_2\text{O} \rightarrow \text{PH}_3 + \text{NaH}_2\text{PO}_2$ [2]
 ii) Define oxidant and reductant giving suitable example. [1]
 - b) Define functional group with example. [2]
11. a) Define functional group of functional group.
 b) Write the characteristics of functional group of sodium extract for detection of [2]
 c) Give the reason for the preparation of sodium extract with necessary heteroelement. [3]
 d) How can you detect nitrogen as heteroelement? Explain it with necessary reaction.

Class: XI 'Science'

Time: 3hrs.

First Term Examination 2080

Set: B

Subject: Chemistry [3011]

F.M.: 75

P.M.: 30

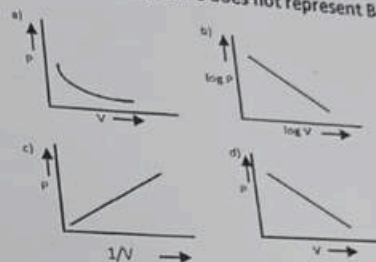
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Attempt all the questions.

GROUP 'A' [MCQ] [11×1=11]

- In which of the following there is transfer of five electrons?
 a) $CrO_3^{2-} \rightarrow Cr^{3+}$ b) $MnO_4^- \rightarrow Mn^{2+}$
 c) $Cr_2O_7^{2-} \rightarrow 2 Cr^{3+}$ d) $CrO_3^{2-} \rightarrow 2 Cr^{3+}$
- The value of x and y in the following redox reaction
 $xCl_2 + 6OH^- \rightarrow ClO_3^- + yCl^- + 3H_2O$ are
 a) x=5, y=3 b) x=2, y=4 c) x=3, y=5 d) x=4, y=2
- When a piece of copper wire is immersed in a solution of silver nitrate, the colour of the solution becomes blue. This is due to
 a) Oxidation of Cu b) Reduction of Cu
 c) Oxidation of Ag^+ d) Formation of soluble complex.
- The mass of oxygen gas whose volume is 320 ml at $17^\circ C$ and 2 atmosphere pressure is
 a) 0.68g b) 0.72g c) 0.86g d) 8.6g
- The volume occupied by 1.008mg of Hydrogen at STP is
 a) 22.4cc b) 11.2cc c) 22.4 lit d) 11.2 lit
- The highest rate of diffusion is found in
 a) NH_3 b) H_2S c) H_2O d) SO_2
- Electron affinity is positive when
 a) O changes into O^- b) O^- changes into O^{2-}
 c) O changes into O^+ d) electron affinity is always negative
- The first organic compound 'Urea' was prepared by
 a) Bezelius b) Wohler c) Kolbe d) Berthelot
- The electronic configuration of an element is $1s^2 2s^2 2p^6 3s^2 3p^3$. The atomic number of the element which is just below this element in the periodic table is
 a) 23 b) 34 c) 36 d) 33

10. Which of the following curve does not represent Boyle's law?



11. Organic cyclic compounds possessing the properties of aliphatic compound is
 a) aromatic b) cyclic c) alicyclic d) homocyclic

GROUP 'B' [8×5=40]

Short Questions.

- Balance the following redox reaction either by oxidation number method or by ion-electron method.
 $Mg + HNO_3 \rightarrow Mg(NO_3)_2 + N_2O + H_2O$
 - Identify the number of acidic and oxidizing HNO_3 in above reaction. [4+1]
- Define oxidation number with example. [1]
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 - Compare the ionic size of F^- , Ne & Na^+ with reason. [3]
- Write short note on "Homologous series & its characteristics." [2]
 - Write differences between organic and inorganic compounds. [2]

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL

Class: XI 'Science'

First-Term Examination 2080

Time: 3hrs.

SET - A

Subject: Physics '1011'

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'A' [11×1=11]

Multiple choice questions

- 1) If energy E , velocity V and force F are taken as fundamental physical quantities, the dimensional formula of mass is
 a) $[F^0 E V^{-2}]$ b) $[F E V^2]$ c) $[F E V]$ d) $[F E^0 V^{-2}]$
- 2) If sum of two-unit vectors is a unit vector, then the magnitude of their difference is
 a) $\sqrt{2}$ b) 2 c) $\sqrt{3}$ d) $\sqrt{5}$
- 3) Which of the following cannot be resultant of the vectors of the magnitude 9 and 3?
 a) 3 b) 6 c) 12 d) 16
- 4) A force of 20 N is applied at an angle of 60° to the horizontal. What is the vertical component of this force?
 a) 20N b) 10N c) 17.32N d) 15N
- 5) The maximum and minimum magnitude of the resultant of two given vectors is 7 units and 1 unit respectively. If these two vectors are at right angles to each other, the magnitude of resultant is
 a) 4 b) 5 c) 6 d) 7
- 6) A certain vector in xy plane has an x-component of 12 m and y-component of 8m. It is then rotated in xy plane so that its x-component is halved. Then its new y component is approximately
 a) 14m b) 10.12m c) 2.14 m d) 13.11m
- 7) An ant moves towards the plane mirror with speed of 3m/s and the mirror is moved towards the ant with the same speed. What is the velocity of ant with respect to its image?
 a) 3 m/s b) 18 m/s c) 9 m/s d) 12 m/s
- 8) The focal length of concave mirror is f and the distance from the object to the principal focus is x . The ratio of size of the image to the size of the object is
 a) $\frac{f+x}{f}$ b) $\frac{x}{f}$ c) $\frac{f}{x}$ d) $\frac{f}{f+x}$
- 9) A point object is placed on the axis of the concave mirror at a distance of 60 cm from the focal point of the mirror. Its image is formed at the point of object then focal length of the mirror is
 a) 60 cm b) 120 cm c) 30 cm d) 15 cm
- 10) Two equally charged, identical metal spheres A and B repel each other with force F . The spheres are fixed with a distance r between them. Third identical but uncharged sphere is brought in contact with A and then placed at midpoint of line joining A and B. The magnitude of net electrostatic force on sphere A is
 a) $3F/2$ b) F c) $2F$ d) $F/2$
- 11) On a new linear scale of thermometer W scale, the freezing and boiling point of water are $39^\circ W$ and $239^\circ W$ respectively. What will be the temperature on the new scale corresponding to the $39^\circ C$ on the Celsius scale
 a) $200^\circ C$ b) $78^\circ C$ c) $139^\circ C$ d) $117^\circ C$

GROUP 'B' [8×5=40]

Short answer questions:

- 12) a) What are the limitations of dimensional analysis? [1]
 b) Check the correctness of the physical relation $p = \frac{3g}{4\pi r G}$ where symbols carry their usual meanings. [2]
 c) Find the value of 20 J on the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units. [2]

energy of C, N, O, F is
 b) $O > F > N > C$
 d) $F > O > C > N$

d) $I^+ < I^- < I$

electrode will not depend on
 a) current intensity b) time of electrolysis
 c) temperature of the both d) electrochemical equivalent of ions
 11. The oxide which cannot act as a reducing agent is
 a) NO_2 b) CO_2 c) SO_2 d) ClO_2

GROUP 'B' [5 × 5 = 25]

Short questions.

- 1) What is **Bus**? Explain all the types of bus with a figure. [1+4]
- 2) What is **Hard disk drive**? Explain all the parts of a hard disk drive with a figure. [1+4]

OR

Write down 5 differences between

- a) **SRAM** and **DRAM** [2.5]
- b) **CRT** AND **LCD** monitors [2.5]

- 3) What is **spreadsheet**? Write down some of the essential features of spreadsheet with examples. Explain relative cell referencing and absolute cell referencing in brief with appropriate table and formula. [1+1+3]

- 4) Explain **OCR**, **OMR**, **MICR**, **BCR** and **SCANNER**.

- 5) What is **mail merge**? Write down the steps to insert mail merge. [1+4]

GROUP 'C' [2 × 8 = 16]

Long questions.

- 1) Explain all the types of **memory** in memory hierarchy in detail. [8]
- 2) Explain the types of **classification** of computers on the basis of size, speed, storage capacity. [8]

Best of Luck!

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Best of Luck!

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2081

SET-A

Subject: Chemistry [3011]

Time: 25 Mins.

Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Multiple Choice Questions.

GROUP 'A' [MCQ] [11×1=11]

The rate of diffusion of which of the following gas is 4 times as compared to oxygen?

- a) H_2 c) NO_2 d) CO_2

The density of NH_3 at $30^\circ C$ and 5 atm pressure is

- a) 2.30 gm/lit b) 2.43 gm/lit c) 3.42 gm/lit d) 17 gm/lit

If atmospheric air contains 20% oxygen by volume, partial pressure of O_2 gas is

- a) 152 mmHg b) 760 mmHg c) 380 mmHg d) 680 mmHg

The increase in volume that will take place when 200 ml of gas at 300K is heated to 360K at constant pressure is

- a) 200 ml b) 160 ml c) 40 ml

In the following equation, the value of 'X' is

- $S_4O_6^{2-} \longrightarrow S_4O_6^{2-} + X$
 a) $3e^-$ b) $4e^-$ c) $5e^-$

In the electrolysis of fused salt, the amount of the substance deposited on the cathode will not depend on

- a) current intensity b) time of electrolysis
 c) temperature of the both d) electrochemical equivalent of ions

The oxide which cannot act as a reducing agent is

- a) CO_2 b) SO_2 c) ClO_2

The number of moles of Faraday needed to reduce a mole of Fe^{3+} to Fe^{2+} are

- a) 2 b) 3 c) 4

The most stable allotrope of sulphur is

- a) monoclinic b) plastic c) milk of sulphur d) rhombic

The correct order of size is

- a) $I^+ < I^- < I$ b) $I^- < I < I^+$ c) $I^+ < I < I^-$ d) $I^- < I^+ < I$

The correct order of ionization energy of C, N, O, F is

- a) $N > O > C$ b) $O > F > N > C$
 c) $N > O > C$ d) $F > O > C > N$

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2081

SET-B

Subject: Chemistry [3011]

Class: XI 'Science'

Time: 25 Mins.

Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Multiple Choice Questions.

GROUP 'A' [MCQ] [11×1=11]

1. The correct order of ionization energy of C, N, O, F is

- a) $F > N > O > C$ b) $O > F > N > C$
 c) $F > O > N > C$ d) $F > O > C > N$

2. The number of moles of Faraday needed to reduce a mole of Fe^{3+} to Fe^{2+} are

- a) 1 b) 2 c) 3 d) 4

3. The density of NH_3 at $30^\circ C$ and 5 atm pressure is

- a) 2.30 gm/lit b) 2.43 gm/lit c) 3.42 gm/lit d) 17 gm/lit

4. The correct order of size is

- a) $I^+ < I^- < I$ b) $I^- < I < I^+$ c) $I^+ < I < I^-$ d) $I^- < I^+ < I$

5. The rate of diffusion of which of the following gas is 4 times as compared to oxygen?

- a) N_2 b) H_2 c) NO_2 d) CO_2

6. The increase in volume that will take place when 200 ml of gas at 300K is heated to 360K at constant pressure is

- a) 240 ml b) 200 ml c) 160 ml d) 40 ml

7. The most stable allotrope of sulphur is

- a) monoclinic b) plastic c) milk of sulphur d) rhombic

8. If atmospheric air contains 20% oxygen by volume, partial pressure of O_2 gas is

- a) 152 mmHg b) 760 mmHg c) 380 mmHg d) 680 mmHg

9. In the following equation, the value of 'X' is

- $2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + X$
 a) $2e^-$ b) $3e^-$ c) $4e^-$ d) $5e^-$

10. In the electrolysis of fused salt, the amount of the substance deposited on the cathode will not depend on

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 c) temperature of the both d) electrochemical equivalent of ions

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